ANSWER KEY & MARKING SCHEME

Class IX – Structure of Atom

Maximum Marks: 40

Case Study 1 (1 Mark)

Answer: Electron

Marking:

• Correct answer – **1**

Case Study 2 (1 Mark)

Answer: Dalton's atomic model

Marking:

• Correct identification – **1**

Case Study 3 (2 Marks)

a) **Rutherford – 1**

b) Atom has a **mostly empty space with a dense nucleus** – **1**

Case Study 4 (2 Marks)

a) Atomic number = **11 – 1**

b) Mass number = **23 – 1**

Case Study 5 (2 Marks)

a) **Cation – 1**

b) Atom becomes **positively charged – 1**

Case Study 6 (2 Marks)

a) **Isobars – 1**

b) Example: **$^{14}_{6}C$ and $^{14}_{7}N$ – 1**

Case Study 7 (2 Marks)

a) **Isotopes – 1**

b) Same **electronic configuration / same atomic number – 1**

Case Study 8 (2 Marks)

a) **Sodium (Na) – 1**

b) **Highly reactive** due to **1 valence electron – 1**

Case Study 9 (2 Marks)

a) **Rutherford's α-particle scattering experiment – 1**

b) **Nucleus is very small and dense – 1**

Case Study 10 (2 Marks)

a) Electrons = **17 – 1**

b) Valency = **1** (needs 1 electron to complete octet) – **1**

Case Study 11 (3 Marks – Tough)

a) **Chloride ion (Cl^-)** – **1**

b) Electronic configuration: **2, 8, 8** – **1**

c) **Anion** – **1**

Case Study 12 (3 Marks – Tough)

a)

• P (10): **2, 8** – ½

• Q (11): **2, 8, 1** – ½

• R (12): **2, 8, 2** – ½

b) Noble gas: **P (Neon)** – ½

c) Most reactive: **Q (Sodium)** due to **1 valence electron** – **1**

Case Study 13 (3 Marks – Tough)

a) Neutrons = 24 − 12 = **12** – **1**

b) Electronic configuration: **2, 8, 2** – **1**

c) Valency = **2** – **1**

Case Study 14 (3 Marks – HOTS)

a) **A^{2+}** has more protons – **1**

b) Loss of electrons increases effective nuclear charge – **1**

c) **Ionic (electrovalent) bond** – **1**

Case Study 15 (3 Marks – Higher Order)

a) Electronic configuration: **2, 8, 8, 2** – **1**

b) Ion formed: **X^{2+} (Ca^{2+})** – **1**

c) Oxide formula: **CaO** – **1**

Case Study 16 (3 Marks – HOTS)

a) Atomic number = **16** – **1**

b) Element: **Sulphur (S)** – **1**

c) Group: **16 (VI A)** – **1**

Case Study 17 (3 Marks – HOTS)

a) Gains **1 electron** – **1**

b) Ion formed: **X^-** – **1**

c) To complete **octet** – **1**

Case Study 18 (3 Marks – Conceptual)

a) Failed to explain **nucleus / α-scattering** – **1**

b) **Rutherford's experiment** – **1**

c) **Rutherford's atomic model** – **1**

Case Study 19 (3 Marks – Numerical Logic)

a) Atomic number = **20** – **1**

b) Electronic configuration: **2, 8, 8, 2** – **1**

c) Element: **Calcium (Ca)** – **1**

Case Study 20 (3 Marks – Competitive)

a) **Isotopes** – **1**

b) Same **atomic number / same electrons** – **1**

c) **Chlorine** – **1**